

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Engineered and deployed **AI model serving architectures** on AWS EC2 instances, achieving a **30% reduction in average response latency** for real-time inference tasks.
- Developed Python scripts to automate the generation of **performance metrics reports** for inference pipelines, tracking throughput, memory consumption, and error rates across development and production environments.
- Designed and implemented **controlled inference tests** in simulated environments, comparing model performance against established benchmarks and documenting iterative results for low-resource devices.
- Containerized **inference pipelines** using Docker and orchestrated deployments on Kubernetes (EKS), ensuring efficient resource utilization and scalability.
- Analyzed computational efficiency and diagnosed bottlenecks in serving pipelines, addressing suboptimal batch processing and high memory usage to optimize infrastructure for resource-constrained systems.
- Collaborated with cross-functional teams to integrate optimized serving and inference frameworks into production pipelines for edge applications, defining success metrics such as improved real-world performance and low error rates.
- Wrote **production runbooks** and internal documentation for the deployment and monitoring of AI inference services, reducing incident resolution time by approximately **20%**.

SKILLS

AI/ML – Model Serving, Inference Optimization, Kernel Optimization, Compute Shaders, Metal Shading Language (MSL), Multi-modal Architectures, AI R&D;

Languages – Python, Java, SQL, Metal Shading Language (MSL)

Cloud & Containerization – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, EKS

Data & Analytics – Pandas, NumPy, Scikit-learn, Data Warehousing, ETL, Data Modelling

DevOps & Tools – Jenkins, Git, Jira, Confluence, Splunk, Dynatrace

Backend Frameworks – Spring Boot, REST APIs, Flask, FastAPI

PROJECTS

Optimized Inference Engine for Edge Devices

- Developed custom compute shaders using **Metal Shading Language (MSL)** for efficient parallel processing on resource-constrained edge devices, targeting a **25% improvement in inference throughput**.
- Built Python scripts to automate the extraction of behavioural signals from raw data, creating feature sets for a predictive model and evaluating performance against real-world deployment challenges.
- Containerized the inference pipeline with Docker and deployed it to a Kubernetes cluster, establishing clear performance targets for reduced latency and minimized memory footprint.

AI Model Serving Pipeline with FastAPI

- Engineered a **FastAPI-based inference pipeline** to serve a scikit-learn classification model, processing approximately **150 requests per second** with p95 latency under 200ms.
- Trained a gradient-boosted classifier on a dataset of ~50,000 records to flag suspicious activities, achieving an **AUC of 0.82** on held-out data.
- Designed and implemented comprehensive monitoring and logging for the inference endpoint, tracking key performance indicators and identifying potential bottlenecks in production.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)